Bridging CNN and Transformer With Cross-Attention Fusion Network for Hyperspectral Image Classification

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***Abstract*— Feature representation is crucial for hyperspectral image (HSI) classification. However, existing convolutional neural network (CNN)-based methods are limited by the convolution kernel and only focus on local features, which causes it to ignore the global properties of HSIs. Transformer-based networks can make up for the limitations of CNNs because they emphasize the global features of HSIs. How to combine the advantages of these two networks in feature extraction is of great importance in improving classification accuracy. Therefore, a cross-attention fusion network bridging CNN and Transformer (CAF-Former) is proposed, which can fully utilize the advantages of CNN in local features and Transformer's long time-dependent feature learning for hyperspectral classification. In order to fully explore the local and global information within an HSI, a Dynamic-CNN branch is proposed to effectively encode local features of pixels, while a Gaussian Transformer branch is constructed to accurately model the global features and long-range dependencies. Moreover, in order to fully interact with local and global features, a cross-attention fusion (CAF) module is proposed as a bridge to fuse the features extracted by the two branches. Experiments over several benchmark datasets demonstrate that the proposed CAF-Former significantly outperforms both CNN-based and Transformer-based state-of-the-art networks for HSI classification.**

***Index Terms*— Convolutional neural network (CNN), feature fusion, hyperspectral image (HSI) classification, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) have complex spatial structure and rich spectral information, whose spatial information effectively reflects the structure, shape, and texture of ground objects, and spectral information reflects the physical and chemical characteristics of constituent substances. Compared with traditional remote sensing images, HSIs are composed of hundreds of adjacent spectral channels, which can more accurately classify ground objects [1], [2], [3], [4] and have been widely used in agricultural production, pathological detection, military reconnaissance, environmental governance, and other fields [5], [6], [7], [8]. As an important prerequisite for applying HSI to the above fields, HSI classification refers to class label assignment to each pixel in an image.

In the early research stage of HSI classification, researchers often used traditional methods to extract features, such as kernel support vector machine (SVM) [9], principal component analysis (PCA) [10], K-nearest neighbor (KNN) [11], and sparse representation (SR) [12], [13], [14], [15], etc. For example, Peng et al. [12] proposed a maximum likelihood estimation-based joint SR model, which can dramatically reduce the effect of inhomogeneous neighboring pixels or outliers. Duan et al. [13] constructed a pair of semi-supervised hypergraphs, which effectively fused labeled and unlabeled training information into the embedding process to obtain the nonlinear spectral–spatial features of HSI. However, these traditional classifiers can only extract spectral features or spatial features, but have difficulty in mining deep nonlinear spectral–spatial correlation information and rely too much on prior information.

Compared with traditional classifiers, deep learning (DL)-based methods have been rapidly developed in visual tasks due to their strong fitting and feature extraction capabilities. Through multilayer feature transformation, DL-based methods can extract higher-dimensional and more abstract features from raw data, thus avoiding heavy feature engineering as in traditional methods. The representative DL networks include stacked autoencoders (SAEs) [16], deep belief networks (DBNs) [17], recurrent neural networks (RNNs) [18], and convolutional neural networks (CNNs) [19], [20], [21], [22], [23]. In the above network, CNN [24], [25], [26], [27], [26], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39] architecture captures local information by using local receptive fields, shared parameters, and spatial subsampling, while being able to skillfully combine spectral features with spatial context information in HSI data. Thus, it gradually becomes a research hotspot in HSI classification. For example, Chen et al. [28] introduced a Gabor filter into CNN, which can effectively extract spatial features including edges and textures while preventing overfitting. Liu et al. [29] aggregated spectral and

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spatial features via CNN, while using more representative class features instead of redundant pixel information in the reconstruction process. Mei et al. [38] designed a CNN-based network with location attention and channel attention to fully explore the bidirectional relationship between adjacent band images in HSI. Lu et al. [39] proposed a hybrid network, which combines CNN with graph convolutional network (GCN) and fuses the extracted local and global information for HSI classification. However, due to the convolution kernel limitation of CNNs, the receptive field of CNN-based methods is confined to the square region of the convolution kernel size, which usually makes the network unable to explore global features and model long-range dependencies of HSIs.

With the proposal of vision transformer (VIT) [40], Transformer-based networks have been introduced into computer vision tasks and achieved satisfactory performance, benefiting from the ability of capturing long-range dependencies. The Transformer architecture is a self-attention-based structure that incorporates a multilayer perceptron (MLPS) to model long-range dependencies of tokens in sequences. Therefore, it can be well applied to HSI classification tasks and derives many visual Transformer-based models [41], [42], [43], [44], [45], [46], [47], [48], [49], [50]. For instance, He et al. [45] proposed a bidirectional encoder representation, which is more flexible than CNN with a larger receptive field and stronger generalization ability. Xue et al. [47] introduced a new attention mechanism in Transformer, which relies on sub-blocks obtained by sliding around the central pixel point. However, Transformer-based networks ignores the local features of images, which may degrade their classification performance.

This naturally leads to a question: how to design effective networks to efficiently encode local processing and global interactions? A simple idea is to combine convolution and Transformer [51], [52], [53], [54], [55], [56]. Recent work has demonstrated the advantages of cascading convolutions and Transformer, either using convolutions at the outset or embedding convolutions within each transformer block. Mei et al. [52] designed an efficient CNN-based pixel embedding module that is able to model local components as a complement to the overseparated features extracted from transformers. Zhou et al. [55] proposed a dual-branch convolutional Transformer method with efficient interactive self-attention, which can aggregate local and global spatial–spectral features fully. However, Transformer-based networks ignores the local features of images, which may degrade their classification performance. However, there is a disadvantage in such a serial design, that is, the Transformer block processes the local features obtained through convolution, and models the context information of the local features, essentially ignoring part of the global information within HSIs.

To address the above problems, a novel network named CAF-Former is proposed, which parallelizes CNN and transformer with a cross-attention fusion (CAF) module in between. In this network, the CNN branch is used to extract local features and multiscale features of HSIs, while the Transformer branch is used to fully explore global features of HSIs. The CAF module adaptively fuses features from the two branches to model their correlation. Finally, experiments over several benchmark datasets, including Indian Pines (IP), Yellow River Estuary coastal wetland (YRE), HyRANK-Loukia (HR-L), and WHU-Hi-HanChuan (WHU-HC), are conducted to verify the effectiveness of the proposed CAF-Former for classification of HSIs.

The main contributions are as follows.

1) A novel network that parallelizes CNN and Transformer is proposed, which can effectively combine the advantages of CNN in local features and Transformer's long-time-dependent feature learning, and fully explore the joint spatial–spectral information of HSIs for classification.
2) A CAF module is proposed as a bridge to fuse the local features of pixels and the context dependency of tokens, which can fully interact with local and global information of HSIs.
3) A Dynamic-CNN is proposed to explore both local features and multiscale features by adjusting the expansion rate and kernel size of dilated convolution, and further strengthen feature representation by using Dynamic Relu function.

The remaining sections of this article are organized as follows. First, Section II describes the details of the proposed method. Second, in Section III, a series of experiments and results are illustrated to compare the proposed model with several state-of-the-art methods. Finally, some conclusions of this article are drawn in Section IV.

II. Proposed Method

A. Overall Framework

The proposed network framework is shown in Fig. 1, which consists of a Dynamic-CNN branch, a Gaussian Transformer branch, a CAF module, and a linear classification module. First, the Dynamic-CNN branch is composed of dilated convolution and Dynamic Relu function, in which dilated convolution is used to extract local features and multiscale features by adjusting the expansion rate and kernel size, and Dynamic Relu function is used to improve model representation by adaptively constructing piecewise linear activation functions based on input. Second, the Gaussian Transformer branch is used to model global features and long-range dependencies, while Gaussian absolute position encoding is introduced into the branch to model the relationship between the central domain and neighboring domain. Third, a CAF model is constructed to bridge between the CNN and the transformer branches, which can perform interactive fusion of local features from the CNN and global features from the Transformer. Moreover, each module is stacked many times to build a hierarchical structure, which reduces the number of feature mapping channels in a hierarchical way and further improves the characterization ability of features. Finally, the extracted depth features are linearized into 1-D vectors to complete the classification. The above modules are described in detail as below.

B. Dynamic-CNN Branch

As shown in Fig. 2, the Dynamic-CNN branch includes two 1×1 convolution layers, a dilated convolution layer,

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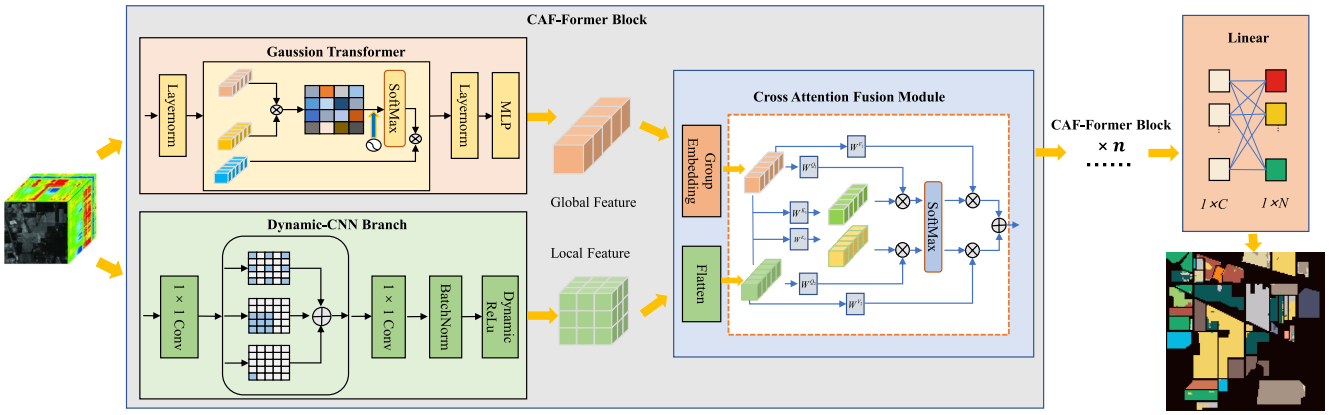


Fig. 1. General schematic of the proposed network model.

and a BatchNorm layer, after which Dynamic Relu function is used to further improve the characterization ability.

For an input $X \in \mathbb{R}^{P\times P\times C}$, where P is the spatial size of input HSI patch and C is the number of HSI bands, it first passes through a 1×1 convolution layer, which is used to split the HSI patch into pixel vectors. Then, these pixel vectors are sent to the dilated convolution layer.

In the dilated convolution layer, the correlations between different pixel vectors are modeled by using a 1×1 convolution and 3×3 dilated convolutions with expansion rates of 0 and 1, respectively. Unlike normal convolution, dilated convolution introduces a hyperparameter of "expansion rate," which defines the spacing of the values between the convolution kernel. When different dilation rates are set, the receptive field will change accordingly, and the pixel vectors processed will be different. In this way, each pixel in the HSI will be fully utilized to model the correlation between similar pixels and the difference between different pixels. Next, 1×1 convolution is used again for segmentation, and the obtained pixel vectors will have more discriminative features. After convolution processing, the feature map is fed into the activation function layer. Traditional activation function formula is as follows:

$$f(x) = \max(0, x) \tag{1}$$

where x represents the feature map and $max(\cdot)$ represents the operation of selecting the maximum value. As shown in Fig. 3(a), it activates each channel in the same way, while HSIs usually have hundreds of channels which are distributed differently. If they are simply processed using traditional activation functions, unique spectral information in different bands may be suppressed. To enhance the inherent characteristics of channels in different bands, the Dynamic Relu function is introduced, as shown in Fig. 3(b), which is a piecewise linear function whose coefficients depend on the input. Let x represent the feature map generated by the convolutional layer, K represent the number of linear functions, and c represent the cth channel, then the Dynamic Relu function can be expressed as

$$y = f_{\theta(x)}(x) = \max_{1\le k\le K} \left\{a_c^k(x)x_c + b_c^k(x)\right\} \tag{2}$$

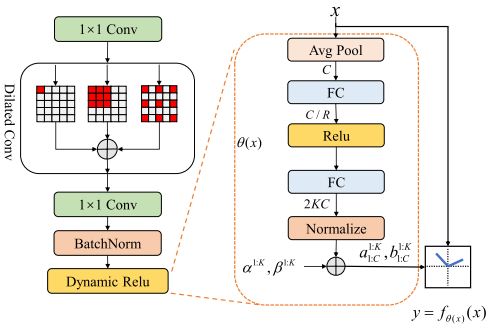


Fig. 2. Structure of Dynamic-CNN and Dynamic Relu activation function.

where $\theta(x)$ is the hyperfunction used to calculate the coefficients a_c^k and b_c^k of different linear functions.

As a learnable function, $\theta(x)$ encodes the global context of the input to achieve the purpose of assigning appropriate coefficients to linear functions. In the proposed model, $\theta(x)$ is generated by a lightweight network, as shown in the right column of Fig. 2. For input x, it is first compressed by global average pooling and then passes through two fully connected layers (with a ReLU between them) and a normalization layer. The outputs $\triangle a_c^k$ and $\triangle b_c^k$, respectively, represent the residual of the coefficients a_c^k and b_c^k, and then their normalization is

$$\left(\Delta a_c^k; \Delta b_c^k\right) = \text{Norm}\left(x'\right) = 2\sigma\left(x'\right) - 1 \tag{3}$$

where $\text{Norm}(\cdot)$ denotes the normalization function, x' represents the feature map after the fully connected layer, and σ is the Sigmoid function. The final output of $\theta(x)$ is computed as the sum of initialization and residual as follows:

$$\begin{aligned} a_c^k(x) &= \alpha^k + \lambda_a \Delta a_c^k(x) \\ b_c^k(x) &= \beta^k + \lambda_b \Delta b_c^k(x) \end{aligned} \tag{4}$$

where α^k and β^k are the initialization of $a_c^k(x)$ and $b_c^k(x)$, respectively. It can be seen from (3) that the Dynamic Relu function's coefficients $a_c^k(x)$ and $b_c^k(x)$ are not only related to the corresponding input channel but also related to other channels, which is beneficial to activate the correlation between channels in different bands while ensuring inherent properties of each channel. The feature map activated by this function possesses part of the global information of HSIs, which increases the compatibility with the features extracted

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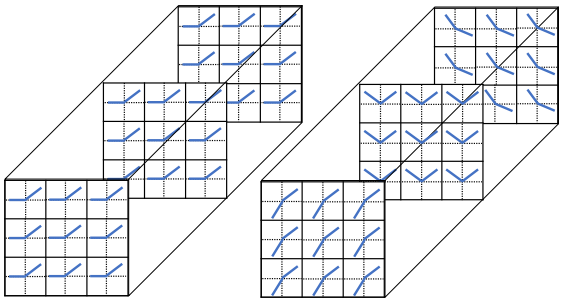


Fig. 3. Piecewise linear function across channels of (a) Relu and (b) Dynamic Relu.

by the Gaussian Transformer branch and is more suitable for subsequent feature fusion. In this method, let $K = 2$, and default values of each hyperparameter are set as follows: $\alpha^1 = 1, \alpha^2 = \beta^1 = \beta^2 = 0, \lambda_a = 1.0$, and $\lambda_b = 0.5$.

C. Gaussian Transformer Branch

The Transformer reconstructs 3-D images into 2-D sequences, which may cause position deviation, so position coding needs to be introduced. The method adopted in this article is based on patch, which is composed of the center pixel and its neighborhood pixel. The distance from the neighboring pixels to the central pixel determines the strength of the correlation between them. When the neighborhood pixels are closer to the central pixel, their spectral information are more similar and the correlation is stronger. On the contrary, the correlation is weaker, even the neighboring pixels are mixed with heterogeneous pixels. To solve the above problems, a Gaussian absolute position encoding is introduced, which uses 2-D Gaussian function to model the relationship between central domains and neighborhoods. In this encoding method, pixels from the same material are enhanced, while heterogeneous pixels and irrelevant noise are suppressed. The structure of the Gaussian Transformer is shown in Fig. 4.

First, the input patch are converted into a 2-D sequence form and sent into multihead self-attention (MHSA). Let s and d denote the length and dimension of the input sequence, respectively. The input 2-D sequence $F \in \mathbb{R}^{s\times d}$ is mapped to three different matrices including query (Q), key (K), and value (V) by linear transformations. They are related as

$$Q_i = \mathrm{FW}_i^Q,\ K_i = \mathrm{FW}_i^K,\ V_i = \mathrm{FW}_i^V \tag{5}$$

where W_i^Q, W_i^K, and $W_i^V \in \mathbb{R}^{d\times d}$ are the learned parameter matrices for the ith head. Since the Transformer model cannot capture marker position information, it is necessary to add encoding modules that represent relative or absolute positions. For each patch, due to the concentrated distribution characteristics of ground objects, the spectral information of pixels closer to the center is more critical than those on the periphery. Therefore, the model adopts a 2-D Gaussian function to represent this spatial relationship of HSI. Let (x, y) denote each pixel's coordinate and (t, t) denote the central pixel's coordinate. Their Gaussian distance can be computed as

$$G(x, y) = e^{-\frac{(x-t)^2+(y-t)^2}{2\sigma^2}}. \tag{6}$$

Then, the Gaussian absolute position encoding is introduced in each head to accomplish self-attention

$$Z_i = \mathrm{Attention}(Q_i, K_i, V_i) = \mathrm{Softmax}\left(Q_i K_i^T + G\right)V_i. \tag{7}$$

The final output of the Gaussian Transformer branch is generated by a projection after concatenating all $Z_i \in \mathbb{R}^{s\times \frac{d}{n}}$, which can be calculated as

$$Z_{\mathrm{Out}} = \mathrm{Concat}\{Z_1, Z_2, \ldots Z_n\}W^O \tag{8}$$

where W^O represents the output projection matrix.

D. Cross-Attention Fusion

Local features are compact adjacency representations of pixels in a local area, while global features refer to the overall attributes of the image, both of which play an important role in classification tasks. However, a single network usually cannot take into account both global and local information. CNNs are limited by the convolution kernel, which makes them unable to model the global characteristics of HSIs. Transformer-based networks convert the input into a 2-D sequence which can model the global information and long-range dependencies, but they cannot capture local features of HSIs. Even changes to Transformer and CNN's architecture cannot overcome the limitations of their own frameworks. In order to take into account both global and local features, the best way is to combine them. However, simply fusing global representations and local features can improve the accuracy of recognition, but also introduce noise side effects. Therefore, how to select the corresponding feature for global representation and local feature fusion is a key issue.

To solve this problem, a CAF module is proposed, which can perform interactive fusion of local features from CNN and global features from Transformer. The structure is shown in Fig. 5.

For the features from the two branches, they are first processed by Flatten and Group Embedding, respectively, to make the dimensions consistent. Group Embedding is actually completed by 1×1 group convolution, which divides the features extracted by the Gaussian Transformer branch into different sub-feature spaces for processing. It aims to give the features local attributes and increase the compatibility with the features extracted by the Dynamic-CNN branch.

Next, the features X_{Conv} extracted from the Dynamic-CNN branch are mapped into the Query matrix Q and Value matrix V, while the values of the Key matrix K are from the Gaussian Transformer branch's features X_{Trans}. The cross-feature concern score is obtained by multiplying Q containing local feature information extracted from convolution with K containing global feature information extracted from Transformer. After the fractional weight is assigned by the activation function layer, it is multiplied by V containing the convolution feature information to carry out further cross-feature fusion interaction. The specific process of the whole process is as follows:

$$X_1 = \mathrm{Softmax}\left(X_{\mathrm{Conv}}W^Q, X_{\mathrm{Trans}}W^K\right)X_{\mathrm{Conv}}W^V \tag{9}$$

where X_1 represents the local features embedded with global information. After this operation, the global properties of HSI

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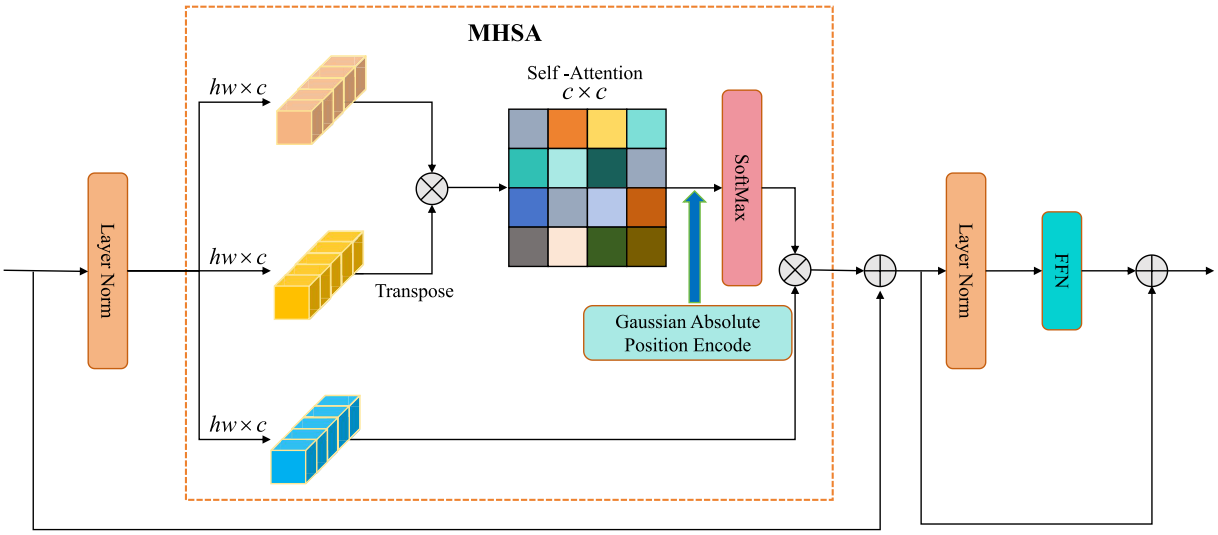


Fig. 4. Structure of Gaussian transformer.

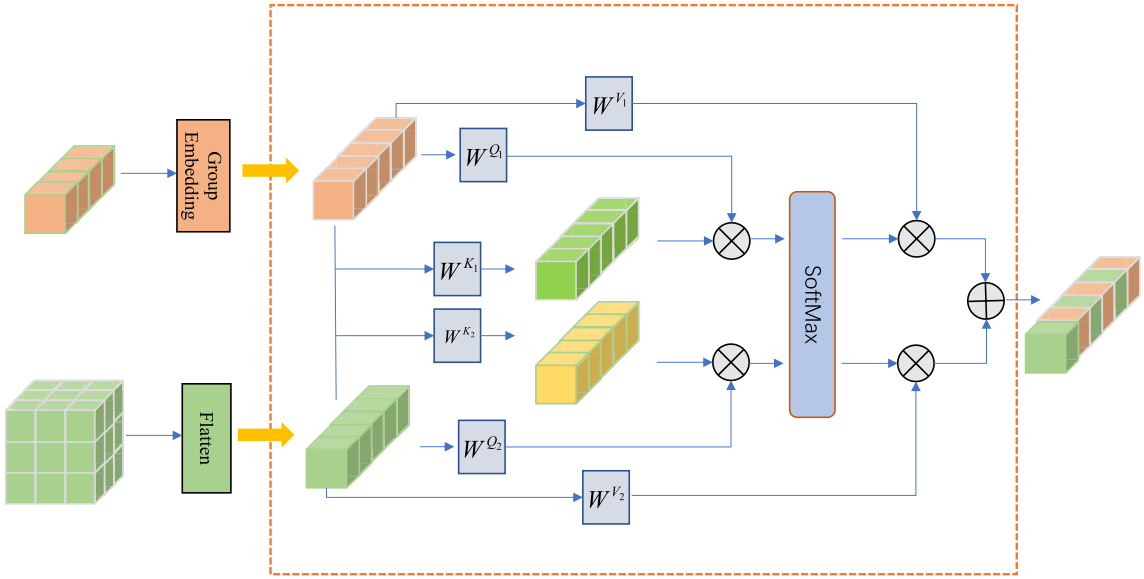


Fig. 5. Structure of CAF module.

are given to the feature map of the Dynamic-CNN branch, allowing it to distinguish useful information from useless noise.

For global features extracted from the Gaussian Transformer branch, the steps to fuse with local features are similar to the above process. Features extracted from the Gaussian Transformer branch are mapped to Query matrix Q and Value matrix V, respectively, while values of Key matrix K are derived from the Dynamic-CNN branch

$$X_2 = \text{Softmax}\left(X_{\text{Trans}} W^Q, X_{\text{Conv}} W^K\right) X_{\text{Trans}} W^V \tag{10}$$

where X_2 represents the global features after giving local information. By embedding local information into feature maps, the Transformer can purposefully screen the global information of HSIs and model pivotal long-range dependencies when performing attention operations.

Finally, the two features are superimposed to complete the CAF as

$$X = X_1 + X_2. \tag{11}$$

III. Experiments

A. Datasets

To evaluate the performance of the proposed method, extensive experiments are conducted over four benchmark datasets, including IP, YRE, HR-L, and WHU-HC.

1) Indian Pines: The IP dataset was imaged by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) in 1992 in Indiana, USA. After removing bands heavily affected by noise, the image has 200 bands, which were used finally as study objects. The data size is 145 × 145. There are 10 249 pixels in total, representing 16 categories.

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TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE IP DATASET

No.	Name	Train.	Val.	Test.
1	Alfalfa	2	3	41
2	Corn-notill	71	72	1285
3	Corn-mintill	42	41	747
4	Corn	12	12	213
5	Grass-pasture	24	24	435
6	Grass-trees	37	36	657
7	Grass-pasture-mowed	2	1	25
8	Hay-windrowed	24	24	430
9	Oats	1	1	18
10	Soybean-notill	48	49	875
11	Soybean-mintill	122	123	2210
12	Soybean-clean	29	30	534
13	Wheat	10	10	185
14	Woods	63	63	1139
15	Buildings-Grass-Trees-Drives	20	19	347
16	Stone-Steel-Towers	5	4	84
	Total	512	512	9225

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE YRE DATASET

No.	Name	Train.	Val.	Test.
1	Reed Marsh	445	444	43550
2	Spartina Alterniflora	307	307	30104
3	Water	2144	2144	210121
4	Tamarix	163	163	15985
5	Tidal Flat Reed	122	123	12001
6	Suacda Salsa	296	296	28997
7	Bare Lake Beach	461	461	45169
8	Bare Land	213	213	20872
	Total	4151	4151	406779

2) Yellow River Estuary Coastal Wetland: The YRE dataset was imaged over the Gaofen-5 satellite over the coastal wetland at the mouth of the Yellow River in Bohai Bay and Laizhou Bay. The image contains 740 × 761 pixels with 296 spectral bands. There are 415 101 pixels in total, representing eight categories.

3) HyRANK-Loukia: The HR-L dataset is one of five satellite HSIs acquired by the EO-1 Hyperion sensor and provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) Science Program. The HR-L image consists of 249 × 945 pixels and 176 spectral bands. In this dataset, a total of 13 503 pixels were manually labeled, including 14 different land covers.

4) WHU-Hi-HanChuan: The WHU-HC dataset was collected by the Headwall NanoHyperspec sensor in HanChuan, Hubei Province, China. The image has a data size of 1217 × 303 and contains 270 spectral bands. The total number of pixels is 257 530, representing 16 types of objects.

Different proportions of samples for each dataset are used for training. For the IP dataset, 5%, 5%, and 90% of the labeled samples are randomly selected for training, validation, and testing, respectively. For the YRE and WHU-HC datasets, 1%, 1%, and 98% of labeled samples are randomly selected for training, validation, and testing, respectively. For the HR-L dataset, 3%, 3%, and 94% of the labeled samples are randomly selected for training, validation, and testing, respectively. The details of dataset partitioning can be seen in Tables I–IV. Moreover, to alleviate the influence of insufficient training data, training samples are further enriched by several data augmentation methods, including vertical flip, horizontal flip, 90° rotation, 180° rotation, and 270° rotation.

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE HR-L DATASET

No.	Name	Train.	Val.	Test.
1	Dense Urban Fabric	8	9	271
2	Mineral Extraction Sites	2	2	63
3	Non Irrigated Arable Land	17	16	509
4	Fruit Trees	3	2	74
5	Olive Groves	42	42	1317
6	Broad-leaved Forest	6	7	210
7	Coniferous Forest	15	15	470
8	Mixed Forest	32	32	1008
9	Dense Sclerophyllous Vegetation	114	114	3565
10	Sparce Sclerophyllous Vegetation	84	84	2635
11	Sparcely Vegetated Areas	12	12	380
12	Rocks and Sand	14	15	458
13	Water	42	42	1309
14	Coastal Water	14	13	424
	Total	405	405	12693

TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE WHU-HC DATASET

No.	Name	Train.	Val.	Test.
1	Strawberry	447	448	43840
2	Cowpea	228	227	22298
3	Soybean	103	103	10081
4	Sorghum	54	53	5246
5	Water spinach	12	12	1176
6	Watermelon	45	46	4442
7	Greens	59	59	5785
8	Trees	179	180	17619
9	Grass	94	95	9280
10	Red roof	105	105	10306
11	Gray roof	169	169	16573
12	Plastic	37	37	3605
13	Bare soil	91	91	8934
14	Road	186	185	18189
15	Bright object	12	11	1113
16	Water	754	754	73893
	Total	2575	2575	252380

B. Experimental Settings

In comparative experiments with other models, for fairness, the batch size and epoch are set to 64 and 300, respectively, to update the training parameters in all frameworks. The

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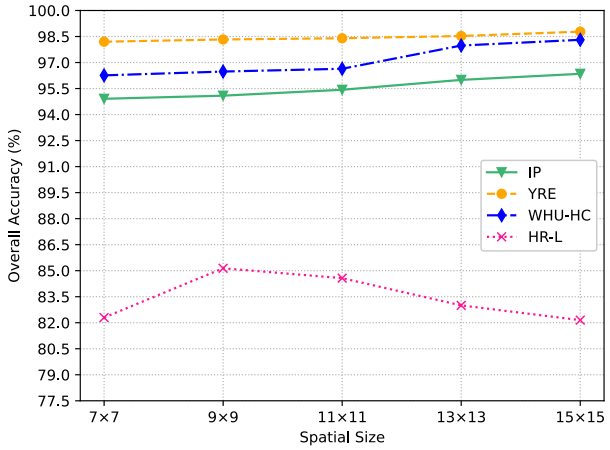


Fig. 6. Overall accuracy (OA) (%) with different spatial sizes on the four datasets.

optimizer and learning rate scheduler are set the same as the original to expect the performance of the other models in the original paper. For the proposed method, the stochastic gradient decent (SGD) optimizer is used with a momentum of 0.9 and a weight decay of 0.0001 to update the training parameters, while setting the learning rate to a constant value of 0.001. The NVIDIA GeForce RTX 3080 GPU with 10 GB of memory is used to accelerate the experiments.

Overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are selected to evaluate the final classification results. OA refers to the ratio of the number of correctly classified pixels to the number of all pixels. AA corresponds to the sum average of the classification accuracy of each single category in all categories, which reflects the overall situation of different categories and can measure the stability of different categories. Kappa coefficient is an indicator used for consistency tests and can also be used to measure the effect of classification, i.e., the so-called consistency is the consistency between model prediction and actual classification results.

To reduce the effects of experimental randomness, the labeled samples are randomly split five times for five rounds of experiments and report their mean and standard deviation. Moreover, color maps are generated to qualitatively compare the results obtained by different methods.

C. Impact of the Input Spatial Size

The size of the patch determines how much spatial information around a pixel is used to classify that pixel. A smaller patch size may make the spatial information captured by the model insufficient, while a larger size may result in too many heterogeneous pixels for degraded classification accuracy. Therefore, selecting an appropriate patch size plays a crucial role. In this part of the experiments, a set of spatial sizes $\{7, 9, 11, 13, 15\}$ are considered to explore their impact on classification results, where the results are shown in Fig. 6.

It can be seen from Fig. 6 that for IP, YRE, and WHU-HC datasets, the classification accuracy gradually increases as the patch size becomes larger. This is because the distribution of features in the three datasets is regular, and the same pixels are gathered together. As the patch size gradually increases, more pixels from the same material are used for classification, which makes the features extracted by the network more discriminative. However, in the HR-L dataset, the best classification results are achieved when the patch size is 9×9. Since the distribution of various ground objects in the HR-L dataset is relatively scattered, as the size increases, heterogeneous pixels and noise are mixed in the patch, which affects the extraction of representative features of various ground objects by the network.

TABLE V
EFFECT OF DIFFERENT NUMBER OF CAF-FORMER BLOCKS ON OA

Number	IP	YRE	HR-L	WHU-HC
1	92.78	97.29	79.71	94.80
2	93.73	97.22	79.86	95.98
3	94.91	**98.20**	**83.08**	**96.26**
4	**94.98**	97.58	81.26	95.69
5	94.75	97.36	81.14	95.71

D. Impact of the Number of CAF-Former Blocks Modules

The CAF-Former block can be superimposed with multiple layers to build a hierarchical network, and the number of superimposed layers determines the depth of network mining features. In order to find an optimal number, a set of comparative experiments with Block numbers ranging into $\{1, 2, 3, 4, 5\}$ are set up. Considering model capacity, when the number of network layers is large, if the spectral dimension remains unchanged, the complexity of the model is greatly increased, which may cause overfitting. For this reason, the spectral dimension is gradually halved as the number of layers deepens. While preventing overfitting, it can also smooth the spectral features and fully mine the advanced features (semantic features, etc.) in HSI.

The experimental results are shown in Table V. On the IP, YRE, and HR-L datasets, the network achieves the best performance when the number of CAF-Former Blocks is three. On the WHU-HC dataset, the network containing four CAF-Former Blocks produces the best performance, but not much advantage over that of using three layers. It can be concluded that the classification results are poor when the network is too deep or too shallow. This is because high-level semantic features cannot be extracted when the number of network layers is relatively shallow. However, as the number of network layers deepens, over-smoothing or over-compression will occur, resulting in a decrease in the degree of discrimination between different nodes and the feature expression of the nodes themselves. Therefore, three is the optimal number of CAF-Former Blocks in the experiments.

E. Comparison With State-of-the-Art Methods

Several state-of-the-art HSI classification methods are adopted for comparison with the CAF-Former, including a deep feature fusion network (DFFN) [57], a residual spectral spatial attention network (RSSAN) [58], an attention-based

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TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE IP DATASET WITH 5% TRAINING SAMPLES

Class	DFFN [57]	RSSAN [58]	AB-LSTM [38]	SpectralFormer [41]	SSFTT [46]	GAHT [52]	MCAL [59]	**CAF-Former**
1	57.07 ± 22.44	32.20 ± 23.26	14.15 ± 27.09	24.39 ± 7.07	55.28 ± 20.15	52.85 ± 24.09	67.32 ± 18.29	**69.11 ± 29.96**
2	89.87 ± 4.92	88.14 ± 3.61	77.04 ± 4.07	67.89 ± 5.26	92.45 ± 0.64	91.62 ± 2.90	**93.70 ± 2.93**	92.94 ± 1.33
3	**96.65 ± 1.10**	83.59 ± 6.42	51.83 ± 25.71	64.58 ± 6.24	94.42 ± 2.87	93.53 ± 1.20	95.15 ± 2.59	96.03 ± 1.38
4	84.32 ± 7.46	72.58 ± 13.83	26.67 ± 21.79	49.01 ± 4.06	88.73 ± 1.01	89.36 ± 4.21	89.30 ± 2.53	**90.92 ± 2.69**
5	92.18 ± 2.93	89.79 ± 3.76	80.46 ± 6.82	74.30 ± 3.99	**94.10 ± 2.18**	92.18 ± 3.26	92.92 ± 3.10	93.03 ± 3.69
6	98.39 ± 0.76	97.20 ± 2.16	95.31 ± 1.63	91.60 ± 3.85	99.75 ± 0.36	99.09 ± 0.65	**99.73 ± 0.28**	99.70 ± 0.33
7	89.60 ± 18.86	56.80 ± 28.67	6.40 ± 12.80	43.20 ± 24.45	90.76 ± 13.20	93.33 ± 9.43	94.40 ± 9.33	**98.67 ± 1.89**
8	98.74 ± 1.48	98.00 ± 1.90	90.56 ± 7.75	93.95 ± 4.69	99.46 ± 0.77	99.46 ± 0.77	99.67 ± 0.65	**99.92 ± 0.11**
9	56.67 ± 24.44	21.11 ± 8.89	0.00 ± 0.00	31.11 ± 8.31	55.56 ± 22.68	53.70 ± 13.09	61.11 ± 25.82	**72.22 ± 20.79**
10	91.25 ± 2.74	83.15 ± 4.82	57.12 ± 20.64	75.18 ± 3.84	92.34 ± 1.12	89.49 ± 2.08	93.10 ± 2.88	**95.20 ± 1.32**
11	95.48 ± 0.29	91.87 ± 2.38	79.36 ± 4.4	81.44 ± 2.14	96.32 ± 0.43	94.13 ± 0.73	95.96 ± 0.89	**96.33 ± 0.63**
12	87.72 ± 4.19	76.33 ± 8.77	44.72 ± 18.93	47.87 ± 3.18	88.26 ± 0.84	90.86 ± 4.14	91.35 ± 3.95	**93.63 ± 2.26**
13	99.14 ± 1.00	96.76 ± 1.13	92.86 ± 3.69	97.08 ± 1.16	99.46 ± 0.76	99.82 ± 0.25	99.79 ± 0.22	**99.82 ± 0.25**
14	97.61 ± 1.33	97.03 ± 1.91	95.28 ± 2.16	92.98 ± 3.09	96.87 ± 2.55	97.92 ± 0.69	97.38 ± 1.81	**98.62 ± 1.05**
15	84.90 ± 5.30	75.91 ± 9.57	43.17 ± 13.20	69.91 ± 7.54	86.74 ± 4.70	87.76 ± 5.03	87.78 ± 5.30	**90.59 ± 6.00**
16	96.43 ± 2.92	86.19 ± 11.56	84.29 ± 13.69	81.43 ± 17.73	92.86 ± 7.72	94.44 ± 4.05	**94.76 ± 5.03**	92.06 ± 8.71
OA (%)	93.57 ± 1.16	88.66 ± 1.40	73.46 ± 6.05	76.86 ± 2.35	94.36 ± 0.42	93.58 ± 0.20	94.92 ± 0.57	**95.64 ± 0.35**
AA (%)	88.50 ± 2.41	77.92 ± 3.46	58.70 ± 7.70	68.84 ± 2.03	88.95 ± 1.36	88.72 ± 1.55	90.84 ± 2.34	**92.42 ± 2.72**
$\kappa\times100$	92.67 ± 1.32	87.04 ± 1.62	69.47 ± 7.23	73.59 ± 2.66	93.56 ± 0.48	92.68 ± 0.24	94.20 ± 0.65	**95.02 ± 0.40**

TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE YRE DATASET WITH 1% TRAINING SAMPLES

Class	DFFN [57]	RSSAN [58]	AB-LSTM [38]	SpectralFormer [41]	SSFTT [46]	GAHT [52]	MCAL [59]	**CAF-Former**
1	95.11 ± 0.74	95.80 ± 0.41	93.51 ± 0.54	94.20 ± 0.31	96.01 ± 1.20	96.69 ± 0.67	96.76 ± 0.24	**97.61 ± 0.47**
2	98.14 ± 0.49	98.23 ± 0.51	97.52 ± 0.66	98.01 ± 0.57	98.59 ± 0.72	98.53 ± 0.27	98.53 ± 0.52	**98.90 ± 0.33**
3	99.66 ± 0.08	99.51 ± 0.08	99.20 ± 0.14	99.44 ± 0.14	99.63 ± 0.11	99.73 ± 0.05	99.66 ± 0.06	**99.71 ± 0.15**
4	93.84 ± 1.39	89.36 ± 2.36	87.35 ± 2.38	88.06 ± 2.84	94.60 ± 0.78	93.42 ± 2.13	94.39 ± 0.96	**96.71 ± 0.64**
5	92.95 ± 1.72	90.44 ± 2.22	86.65 ± 1.76	88.71 ± 2.08	93.43 ± 1.23	93.96 ± 3.61	92.45 ± 0.67	**95.80 ± 1.72**
6	89.37 ± 1.85	90.59 ± 1.60	87.44 ± 2.44	89.00 ± 1.69	90.16 ± 0.89	93.14 ± 1.05	90.56 ± 1.13	**93.80 ± 0.63**
7	93.75 ± 0.72	93.45 ± 1.27	93.27 ± 0.95	93.54 ± 1.04	95.06 ± 0.41	94.49 ± 1.06	94.74 ± 0.98	**95.36 ± 0.35**
8	96.15 ± 1.02	96.30 ± 0.94	95.66 ± 0.71	96.10 ± 0.60	97.14 ± 0.53	97.96 ± 0.46	97.22 ± 0.39	**98.00 ± 0.61**
OA (%)	97.07 ± 0.14	96.88 ± 0.09	97.48 ± 0.13	95.95 ± 0.24	96.44 ± 0.28	97.75 ± 0.11	97.53 ± 0.08	**98.20 ± 0.13**
AA (%)	94.87 ± 0.28	94.21 ± 0.37	95.58 ± 0.14	92.57 ± 0.35	93.38 ± 0.44	95.99 ± 0.42	95.54 ± 0.08	**96.99 ± 0.26**
$\kappa\times100$	95.77 ± 0.20	95.50 ± 0.14	94.17 ± 0.35	96.36 ± 0.35	94.86 ± 0.40	96.76 ± 0.16	96.44 ± 0.11	**97.41 ± 0.19**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE HR-L DATASET WITH 3% TRAINING SAMPLES

Class	DFFN [57]	RSSAN [58]	AB-LSTM [38]	SpectralFormer [41]	SSFTT [46]	GAHT [52]	MCAL [59]	**CAF-Former**
1	37.71 ± 17.86	49.37 ± 15.29	35.13 ± 20.10	47.75 ± 9.05	**69.99 ± 2.56**	48.91 ± 2.25	55.35 ± 6.08	64.05 ± 5.67
2	79.68 ± 16.35	61.27 ± 26.00	70.48 ± 11.47	98.41 ± 3.17	83.07 ± 22.83	86.98 ± 15.11	66.98 ± 25.46	**100.00 ± 0.00**
3	80.16 ± 4.35	82.08 ± 2.57	80.04 ± 7.59	66.44 ± 6.55	74.72 ± 3.46	**88.35 ± 5.50**	88.06 ± 6.28	83.00 ± 7.62
4	4.86 ± 7.33	14.59 ± 9.10	0.00 ± 0.00	1.89 ± 2.36	54.95 ± 3.55	15.11 ± 5.14	25.68 ± 13.99	**57.08 ± 3.60**
5	88.93 ± 3.48	90.33 ± 4.14	79.65 ± 6.23	70.90 ± 4.22	83.30 ± 1.87	**93.69 ± 1.76**	90.98 ± 3.55	84.01 ± 4.11
6	27.52 ± 6.41	34.48 ± 5.81	10.95 ± 15.84	13.05 ± 7.07	48.73 ± 9.56	**58.96 ± 14.03**	48.00 ± 10.95	57.65 ± 9.08
7	46.98 ± 9.40	56.77 ± 5.95	45.96 ± 9.57	42.21 ± 5.77	64.26 ± 6.66	62.60 ± 14.00	58.43 ± 8.38	**73.98 ± 0.91**
8	61.75 ± 7.19	70.54 ± 5.84	62.62 ± 4.13	62.28 ± 6.11	70.14 ± 9.45	65.23 ± 4.85	**70.62 ± 5.40**	64.56 ± 4.49
9	82.12 ± 2.17	79.72 ± 2.14	**82.51 ± 2.28**	72.07 ± 4.82	78.88 ± 1.02	78.38 ± 5.22	82.74 ± 2.64	81.37 ± 2.14
10	78.68 ± 4.04	80.88 ± 4.24	81.24 ± 2.48	71.43 ± 5.95	79.95 ± 1.98	81.71 ± 1.06	80.52 ± 2.34	**88.35 ± 1.58**
11	56.89 ± 5.59	59.42 ± 7.69	59.74 ± 7.46	57.11 ± 5.50	69.74 ± 5.48	62.41 ± 5.64	67.26 ± 6.06	**70.48 ± 2.82**
12	90.13 ± 2.82	86.29 ± 5.35	90.66 ± 3.44	91.09 ± 2.89	**93.89 ± 1.54**	91.72 ± 1.06	93.67 ± 1.04	84.84± 10.02
13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	**100.00 ± 0.00**	99.91 ± 0.12	99.91 ± 0.12	99.95 ± 0.09	98.90 ± 0.11	**100.00 ± 0.00**	99.95 ± 0.09	99.84 ± 0.23
OA (%)	78.78 ± 0.87	80.10 ± 0.39	78.18 ± 1.32	71.98 ± 0.98	80.46 ± 0.09	81.11 ± 1.29	82.16 ± 0.34	**83.08 ± 0.87**
AA (%)	66.82 ± 2.08	68.97 ± 1.77	64.21 ± 1.65	63.90 ± 0.47	76.46 ± 1.80	73.86 ± 2.41	74.82 ± 2.16	**79.23 ± 1.07**
$\kappa\times100$	74.54 ± 1.06	76.26 ± 0.44	73.81 ± 1.54	66.54 ± 1.06	76.82 ± 0.17	77.53 ± 1.45	78.72 ± 0.42	**79.84 ± 1.01**

bidirectional long-short-term memory network (AB-LSTM) [38], a transformer-based backbone network (SpectralFormer) [41], a spectral–spatial feature tokenization transformer network (SSFTT) [46], a group-aware hierarchical transformer network (GAHT) [52], and a multiscale and cross-level attention learning network (MCAL) [59]. For fairness, the spatial size of the input in each model is set to 7 × 7. The remaining parameters are consistent with the original set of parameters for each model to reproduce its best performance.

The experimental results are shown in Tables VI–IX. For the IP dataset, the CAF-Former performs much better than other methods, where OA, AA, and KAPPA reach 95.64%, 92.42%, and 95.02%, respectively. Similar results can also be generated for WHU-HC and YRE datasets, where the CAF-Former

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TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-HC DATASET WITH 1% TRAINING SAMPLES

Class	DFFN [57]	RSSAN [58]	AB-LSTM [38]	SpectralFormer [41]	SSFTT [46]	GAHT [52]	MCAL [59]	**CAF-Former**
1	97.29 ± 0.68	97.32 ± 0.65	95.43 ± 0.89	96.30 ± 0.78	97.78 ± 0.47	97.73 ± 0.12	98.31 ± 0.07	**98.43 ± 0.18**
2	95.07 ± 1.55	94.65 ± 1.14	86.55 ± 2.11	84.95 ± 2.43	96.35 ± 0.67	93.24 ± 1.09	**96.58 ± 0.31**	95.82 ± 1.40
3	95.42± 1.26	92.66 ± 2.74	79.27 ± 1.74	91.40 ± 1.62	95.08 ± 2.71	95.54 ± 2.03	95.65 ± 1.57	**97.95 ± 0.73**
4	96.54 ± 1.07	96.30 ± 0.59	90.98 ± 1.46	95.33 ± 0.67	**97.43 ± 1.22**	97.21 ± 1.06	97.17 ± 1.28	97.18 ± 1.01
5	**86.80 ± 7.24**	82.16 ± 5.97	35.73 ± 13.94	66.19 ± 9.42	83.13 ± 4.46	86.60 ± 4.18	76.73 ± 12.76	82.20 ± 5.36
6	64.98 ± 7.25	61.30 ± 4.65	20.32 ± 6.49	41.58 ± 3.59	72.11 ± 0.52	**72.99 ± 4.22**	71.42 ± 1.67	71.45 ± 5.34
7	89.38 ± 2.63	87.11 ± 6.90	77.20 ± 2.07	80.03 ± 2.37	89.88 ± 2.41	88.70 ± 1.39	90.46 ± 0.79	**91.48 ± 2.10**
8	88.61 ± 0.11	89.00 ± 2.06	72.62 ± 3.44	77.98 ± 2.19	93.47 ± 2.38	89.52 ± 2.91	92.29 ± 0.38	**93.98 ± 1.51**
9	91.23 ± 1.49	88.04 ± 2.31	69.05 ± 4.48	71.73 ± 3.68	93.50 ± 0.64	92.52 ± 1.20	93.26 ± 0.62	**94.62 ± 0.33**
10	98.47 ± 0.37	96.86 ± 0.78	94.58 ± 1.01	96.65 ± 0.44	98.04 ± 0.41	**98.78 ± 0.33**	97.29 ± 0.43	98.27 ± 0.82
11	95.81 ± 1.53	**96.96 ± 1.51**	92.89 ± 2.42	95.07 ± 1.35	94.01 ± 0.62	95.61 ± 0.70	96.24 ± 1.06	96.55 ± 1.56
12	78.80 ± 7.46	74.19 ± 8.30	38.04 ± 11.76	55.23 ± 3.87	83.64 ± 3.76	82.84 ± 1.59	85.11 ± 2.14	**88.16 ± 1.53**
13	79.95 ± 2.16	75.37 ± 3.76	62.21 ± 3.00	67.34 ± 2.91	83.21 ± 3.21	79.65 ± 3.15	81.61 ± 3.11	**84.00 ± 3.47**
14	91.93 ± 1.88	92.11 ± 1.08	86.38 ± 0.91	87.90 ± 1.41	93.97 ± 0.83	**95.54 ± 0.32**	93.64 ± 0.48	95.03 ± 0.05
15	79.30 ± 6.66	74.68 ± 7.51	71.48 ± 5.46	62.26 ± 12.18	80.77 ± 1.28	81.80 ± 5.45	83.38 ± 3.37	**86.73 ± 5.39**
16	99.30 ± 0.22	99.49 ± 0.22	99.04 ± 0.58	96.90 ± 2.21	99.59 ± 0.25	99.55 ± 0.07	99.67 ± 0.13	**99.61 ± 0.17**
OA (%)	94.57 ± 0.70	94.03 ± 0.44	87.43 ± 0.60	89.92 ± 0.30	95.61 ± 0.26	95.15 ± 0.12	95.71 ± 0.13	**96.26 ± 0.13**
AA (%)	89.30 ± 1.91	87.39 ± 1.39	73.24 ± 1.65	79.32 ± 0.78	90.75 ± 0.76	90.49 ± 0.20	90.55 ± 0.77	**91.97 ± 0.80**
$\kappa\times100$	93.65 ± 0.81	93.00 ± 0.52	85.26 ± 0.71	88.07 ± 0.35	94.86 ± 0.31	94.32 ± 0.29	94.97 ± 0.15	**95.62 ± 0.16**

was ahead of other models in OA, AA, and KAPPA. For YRE dataset, the CAF-Former has more obvious performance advantages. In addition to the three evaluation indicators, which are higher than other models, the classification accuracy of each class also reaches the highest level. For the HR-L dataset, which has the smallest number of samples and scattered features, the CAF-Former still maintains its excellent performance, which achieves 0.92%–11.10%, 2.77%–15.33%, and 1.12%–13.30% higher than other models in OA, AA, and Kappa, respectively.

The experimental results further confirm our theoretical analysis, that is, CNN-based methods cannot model the global information of HSI while transforming-based methods lack the extraction of local features of HSI. For CNN-based networks such as RSSAN, the results are poor on the three datasets of IP, YRE, and WHU-HC. The reason is that these three datasets are relatively smooth and the same class of pixels is relatively concentrated. However, due to the limitation of the convolution kernel size, the receptive field of CNN can only be limited to the space of the convolution kernel size, and all pixels in the patch cannot be considered, resulting in the inability to extract global information. For the HR-L dataset with relatively scattered pixels of the same type, the transformer-based network such as SpectralFormer yields the worst classification results. Although SpectralFormer can handle global information well, its ability to extract local information is limited, and cannot suppress the interference from other categories of pixels. GAHT also achieved excellent classification results on four datasets but still lagged behind the CAF-Former. The reason is that although GAHT combines CNN and Transformer to improve the extraction of global information and local information, due to its single-branch structure, Transformer blocks in GAHT are modeled on the basis of features extracted from the CNN block, resulting in the inability of GAHT to extract complete global features. However, the CAF-Former adopts a two-branch structure and integrates the features of the two branches in a more compatible way through CAF, which can fully take into account global information and multiscale local information. The experimental results also demonstrate the effectiveness of the combination.

For qualitative comparison, the one with the highest OA value is chosen among the five experiments to visualize the predicted samples of different methods. As shown in Figs. 7–10, on the four benchmark datasets, the proposed CAF-Former provides the results closest to the ground truth with less noise. This further illustrates its superiority.

F. Ablation Experiments

To analyze the contributions of different components in the CAF-Former, ablation experiments are constructed on four datasets. As described in Section III-A, the proposed network consists of three modules: Dynamic-CNN branch, Gaussian Transformer branch, and CAF module. Below the importance of these three modules to the CAF-Former is verified through experiments, respectively.

1) Dynamic Relu: In the Dynamic-CNN branch, the Dynamic Relu is used to improve the representation ability of the network. To demonstrate this, the Dynamic Relu function is replaced with Relu activation function to train on four benchmark datasets. The result is shown in Fig. 11(a).

Experiments have shown that the Dynamic Relu activation function used by the Dynamic-CNN branch can indeed improve the representation ability of the network. This is because the Dynamic Relu activation function adaptively adjusts the slope according to the characteristics of the input, and fully takes into account the influence of each channel.

Additionally, experiments and analyses were conducted on the computational complexity of two activation functions, as illustrated in Table X. Employment of the Dynamic Relu activation function led to an inevitable increase in the model parameter count compared to traditional Relu, but the

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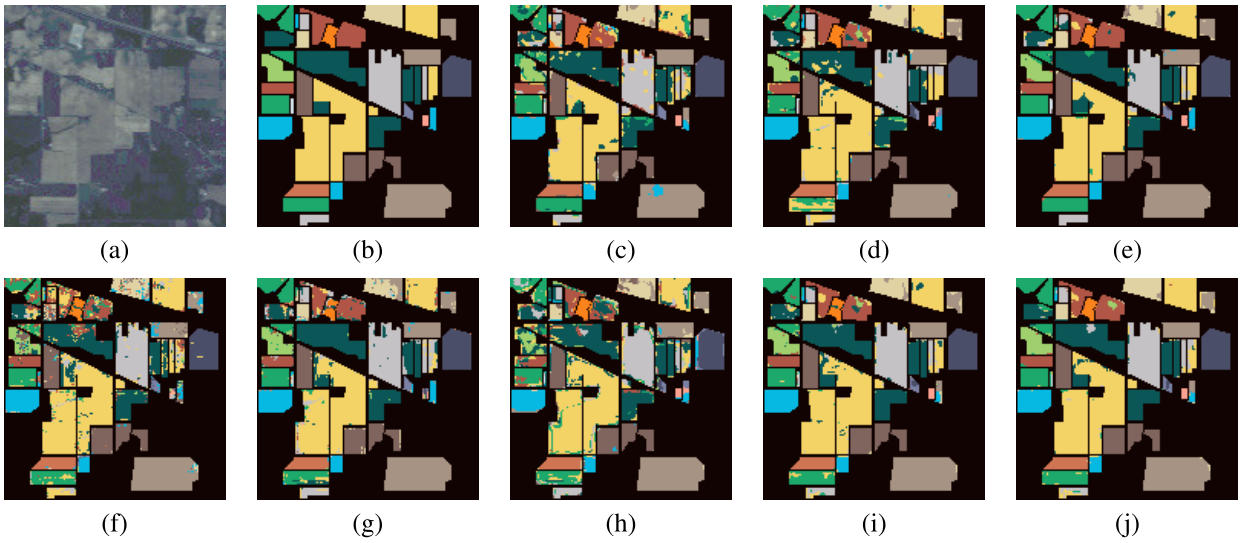


Fig. 7. Classification maps of the IP dataset with 5% training samples. (a) False-color image. (b) Ground truth. (c) DFFN. (d) AB-LSTM. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) GAHT. (i) MCAL. (j) CAF-Former.

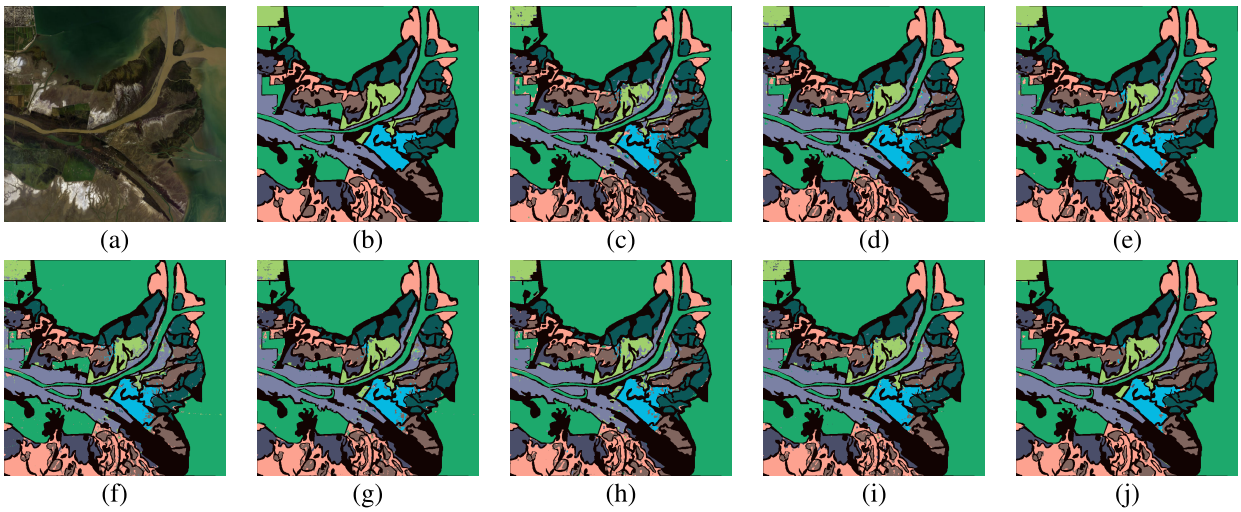


Fig. 8. Classification maps of the YRE dataset with 1% training samples. (a) False-color image. (b) Ground truth. (c) DFFN. (d) AB-LSTM. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) GAHT. (i) MCAL. (j) CAF-Former.

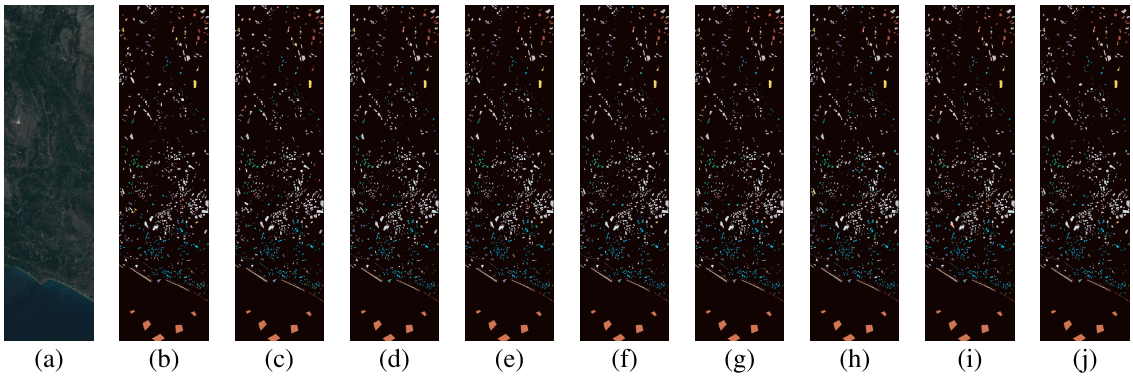


Fig. 9. Classification maps of the HR-L dataset with 3% training samples. (a) False-color image. (b) Ground truth. (c) DFFN. (d) AB-LSTM. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) GAHT. (i) MCAL. (j) CAF-Former.

difference is not significant. For instance, on the IP dataset, the model parameter count increased by 0.11M when using Dynamic Relu, while yielding an improvement of over 2% in OA performance.

2) Gaussian Absolute Position Encoding: In the Gaussian Transformer branch, the Gaussian absolute position encoding is used to correct the influence between neighboring pixels. To show this, two sets of comparative experiments are set up,

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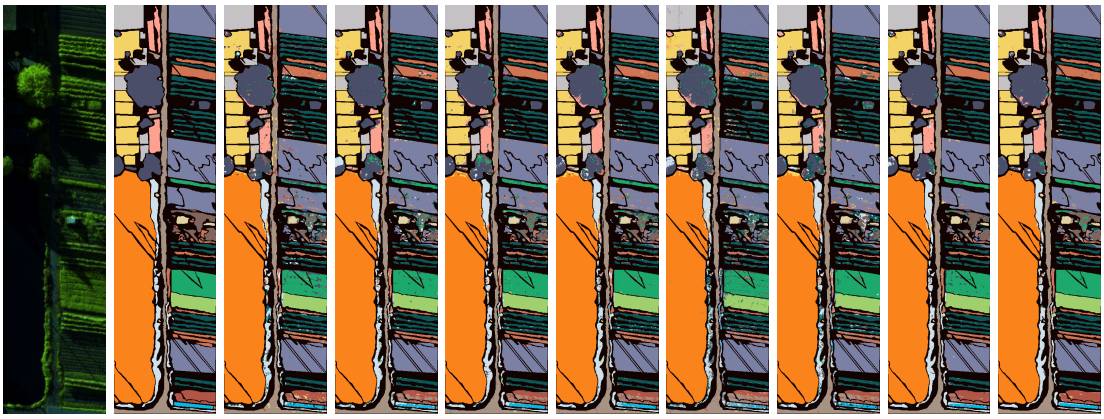


Fig. 10. Classification maps of the WHU-HC dataset with 1% training samples. (a) False-color image. (b) Ground truth. (c) DFFN. (d) AB-LSTM. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) GAHT. (i) MCAL. (j) CAF-Former.

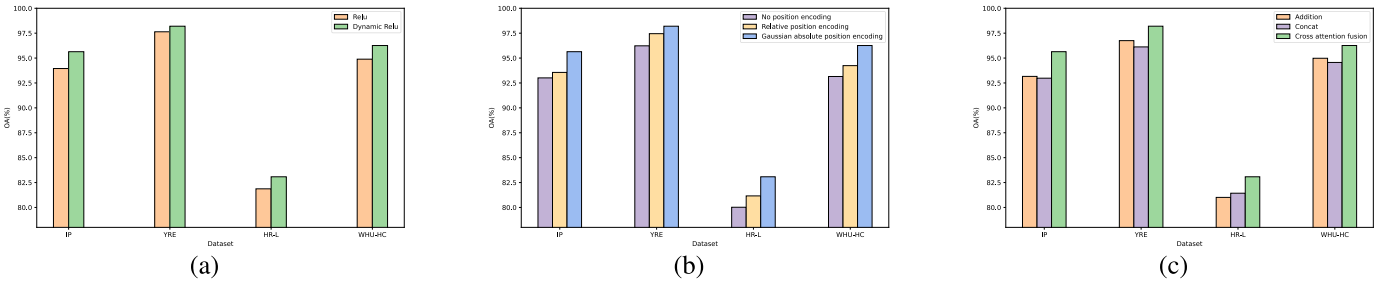


Fig. 11. Ablation experiments. (a) Effect of activation function on OA (%). (b) Effect of position encoding on OA (%). (c) Effect of fusion method on OA (%).

one is to remove the position encoding in Transformer, and the other is to replace the proposed Gaussian absolute position encoding with relative position encoding. The result is shown in Fig. 11(b).

It can be seen from the experimental results that the Transformer without position encoding has the worst result, followed by the addition of relative encoding. This is because the patch used for classification is composed of the central pixel and its neighboring pixels. When the pixels in the neighborhood are closer to the central pixel, the correlation is stronger, otherwise the correlation is weaker, even heterogeneous pixels. The Transformer that lacks position encoding cannot construct this relationship, while relative position encoding is to construct the relationship between pixels, not the relationship between neighboring pixels and intermediate pixels. Gaussian absolute position coding encodes each pixel by calculating the Gaussian distance between the central pixel and other pixels, which can effectively model the relationship between the central domain and the neighborhood of input patch and capture the global features of HSIs. In addition, compared with traditional sine and cosine functions, the Gaussian function is a continuously smooth function, which provides a smoother position representation and helps the model learn continuous changes in position information.

3) Cross-Attention Fusion: The CAF is constructed to encode both local processing and global interaction. In order to fully verify the validity of this module, two sets of comparative experiments are set, one is to change the feature fusion mode to direct addition, and the other is to change the feature fusion mode to concat.

TABLE X
PARAMETERS OF TWO ACTIVATION FUNCTIONS (M)

Dataset	IP	YRE	HR-L	WHU-HC
Relu	2.17	2.21	2.20	2.16
Dynamic Relu	2.28	2.32	2.30	2.27

The experimental results are shown in Fig. 11(c), from which it can be seen that the classification accuracy is highest when the fusion mode is the proposed CAF. This is because the fusion method of adding features does not take into account the different degrees of focus corresponding to different parts of the feature map, while the fusion method of concat is the direct stitching of two groups of features in the spectral dimension, without considering the correlation between them. The proposed CAF is a weighted fusion of different parts of the feature map, which can enhance useful information, suppress redundant noise, and fully model the correlation between features from the two branches.

G. Robustness Evaluation

To evaluate the robustness of the proposed network, our method is tested with other competitors along with the increasing number of training samples. Specifically, 5%, 10%, 15%, and 20% labeled samples were randomly chosen as training data for IP dataset, while 2%, 3%, 4%, and 5% for HR-L dataset, and 0.25%, 0.5%, 1%, and 2% labeled samples were

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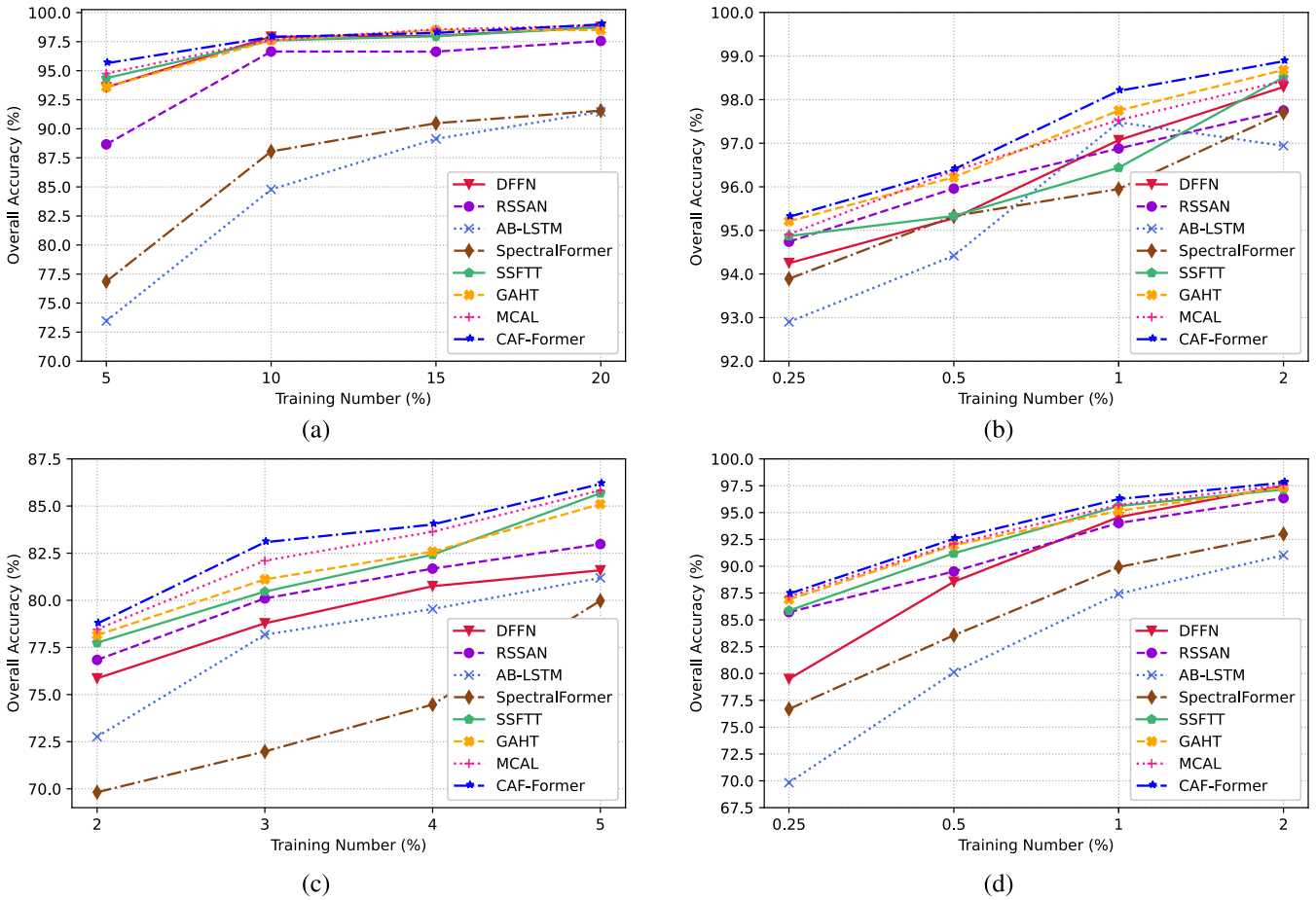


Fig. 12. Classification results versus different percentages of training samples on the four datasets. (a) IP. (b) YRE. (c) HR-L. (d) WHU-HC.

randomly chosen as training data for YRE and WHU-HC datasets.

Fig. 12 shows the OA results of different classifiers. It can be observed that as the number of training samples increases, the OA performance of CAF-Former on the four benchmark datasets increases accordingly, indicating its excellent robustness. Additionally, CAF-Former significantly outperforms other methods in a majority of cases. For instance, when the proportion of labeled samples in the IP dataset is 5%, CAF-Former sustains OA exceeding 95%, whereas the performance of SpectralFormer and AB-LSTM falls below 80%. Similar phenomena also occur on the YRE and WHU-HC datasets, where CAF-Former achieves OA surpassing 95% and 87.5%, respectively, even with a limited number of samples. For the most challenging HR-L dataset, although OA only reaches about 79% in the case of 2% training samples, the performance of CAF-Former is still better than other methods. This is because CAF-Former can make full use of CNN's advantages in local features and Transformer's advantages in long-term reliance feature learning, which enables it to fully capture the features of samples and enhance the robustness of the model.

IV. Conclusion

This article proposes a novel HSI classification method named CAF-Former, which parallelizes Transformer and CNN with CAF module in between. The Dynamic-CNN branch extracts local features in HSIs by combining dilated convolution with Dynamic Relu activation function, and the Gaussian Transformer branch is constructed to explore global information of pixels by using MHSA and model the relationship between the center pixel and neighborhood pixels by introducing the Gaussian absolute position encoding. To efficiently encode local processing and global interactions, a CAF module is proposed to bridge between the CNN and the transformer branches, which can perform interactive fusion of local features from the CNN and global features from the Transformer. A series of experiments over four benchmark datasets demonstrate that the proposed CAF-Former obviously outperforms several state-of-the-art networks for HSI classification. Nevertheless, compared with single branch network, the number of CAF-Former's parameters will inevitably increase. In future research, the network structure deserves to be further optimized to reduce computational complexity.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] G. Zhang et al., "Spectral variability augmented sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 6014605.

[3] Z. Wu, J. Sun, Y. Zhang, Z. Wei, and J. Chanussot, "Recent developments in parallel and distributed computing for remotely sensed big data processing," *Proc. IEEE*, vol. 109, no. 8, pp. 1282–1305, Aug. 2021.

[4] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:16:07 UTC from IEEE Xplore. Restrictions apply.

[5] L. Sun, F. Wu, T. Zhan, W. Liu, J. Wang, and B. Jeon, "Weighted nonlocal low-rank tensor decomposition method for sparse unmixing of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1174–1188, Mar. 2020.

[6] X. Zhang, L. Liu, X. Chen, Y. Gao, and M. Jiang, "Automatically monitoring impervious surfaces using spectral generalization and time series Landsat imagery from 1985 to 2020 in the Yangtze River Delta," *J. Remote Sens.*, vol. 2021, pp. 1–16, Jan. 2021.

[7] M. Dalponte, H. O. Ørka, T. Gobakken, D. Gianelle, and E. Næsset, "Tree species classification in boreal forests with hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2632–2645, May 2013.

[8] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 51–64, May 2022.

[9] G. Camps-Valls, L. Gomez-Chova, J. Munoz-Mari, J. Vila-Frances, and J. Calpe-Maravilla, "Composite kernels for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 93–97, Jan. 2006.

[10] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[11] E. Blanzieri and F. Melgani, "Nearest neighbor classification of remote sensing images with the maximal margin principle," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1804–1811, Jun. 2008.

[12] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[13] Y. Duan, H. Huang, and T. Wang, "Semisupervised feature extraction of hyperspectral image using nonlinear geodesic sparse hypergraphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515115.

[14] Y. Duan, H. Huang, and Y. Tang, "Local constraint-based sparse manifold hypergraph learning for dimensionality reduction of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 613–628, Jan. 2021.

[15] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[16] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[17] T. Li, J. Zhang, and Y. Zhang, "Classification of hyperspectral image based on deep belief networks," in *Proc. IEEE Int. Conf. Image Process.*, Oct. 2014, pp. 5132–5136.

[18] Q. Liu, F. Zhou, R. Hang, and X. Yuan, "Bidirectional-convolutional LSTM based spectral–spatial feature learning for hyperspectral image classification," *Remote Sens.*, vol. 9, no. 12, p. 1330, Dec. 2017.

[19] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, p. 436, 2015.

[20] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[21] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[22] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[23] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[24] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, pp. 1–12, Jul. 2015.

[25] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, Jun. 2015.

[26] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[27] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[28] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[29] Q. Liu, Z. Wu, X. Jia, Y. Xu, and Z. Wei, "From local to global: Class feature fused fully convolutional network for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 24, p. 5043, Dec. 2021.

[30] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.

[31] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[32] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[33] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502012.

[34] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[35] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.

[36] M. Liang, H. Wang, X. Yu, Z. Meng, J. Yi, and L. Jiao, "Lightweight multilevel feature fusion network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 1, p. 79, Dec. 2021.

[37] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[38] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[39] Y. Lu, S. Mei, F. Xu, M. Ma, and X. Wang, "Separable deep graph convolutional network integrated with CNN and prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516216.

[40] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–21.

[41] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[42] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.

[43] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[44] X. Tan, K. Gao, B. Liu, Y. Fu, and L. Kang, "Deep global–local transformer network combined with extended morphological profiles for hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 15, no. 3, Sep. 2021, Art. no. 038509.

[45] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[46] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[47] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.

[48] W. Zhou, S.-I. Kamata, H. Wang, and X. Xue, "Multiscanning-based RNN—Transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512319.

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[49] A. B. Ramathilagam, S. Natarajan, and A. Kumar, "TransCropNet: A multichannel transformer with feature-level fusion for crop classification in agricultural smallholdings using sentinel images," *J. Appl. Remote Sens.*, vol. 17, no. 2, Apr. 2023, Art. no. 024501.
[50] Z. Huang, H. Tang, Y. Li, and W. Xie, "HFC-SST: Improved spatial–spectral transformer for hyperspectral few-shot classification," *J. Appl. Remote Sens.*, vol. 17, no. 2, May 2023, Art. no. 026509.
[51] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 1–15, Nov. 2022.
[52] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.
[53] C. Ma et al., "Light self-Gaussian-attention vision transformer for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023, Art. no. 5015712.
[54] Y. Fang, Q. Ye, L. Sun, Y. Zheng, and Z. Wu, "Multiattention joint convolution feature representation with lightweight transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513814.
[55] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.
[56] Z. Qiu, J. Xu, J. Peng, and W. Sun, "Cross-channel dynamic spatial–spectral fusion transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528112.
[57] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.
[58] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.
[59] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.



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